

Mission Overview:

Sardonyx will demonstrate and characterize the performance of two enabling technologies for **in-space assembly and manufacture**: **electroadhesion to unprepared surfaces** using eTAP and **agile close-proximity maneuvering of a space vehicle near an uncontrolled space object** using the Alice reinforcement-learning controller.



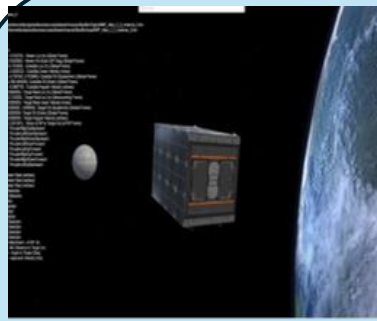
Test and measure a spacecraft's ability to detect, maneuver relative to, and approach very close to a passive, free-flying object



Test and measure the eTAP's ability to adhere to a passive, free-flying object



Test and measure the Alice reinforcement-learned trained controller's ability to autonomously maneuver a spacecraft relative to a passive, free-flying object

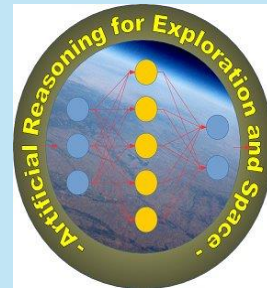


Experiment : Learning to Fly

Motivation: Reinforcement learning could be an effective way for navigating near irregular, gently-tumbling objects

Goals:

- Collect in-flight data of relative navigation
- Maneuver a spacecraft using a reinforcement-learning system



How Close is Too Close?

Proximity: ~200 meters

Close-Proximity: ~10 meters

Sardonyx: even closer!

Constraints = Creativity

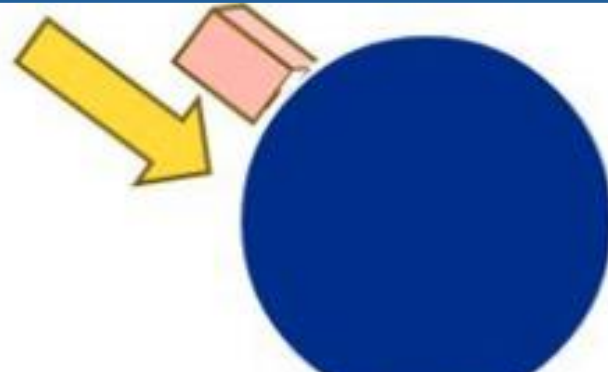
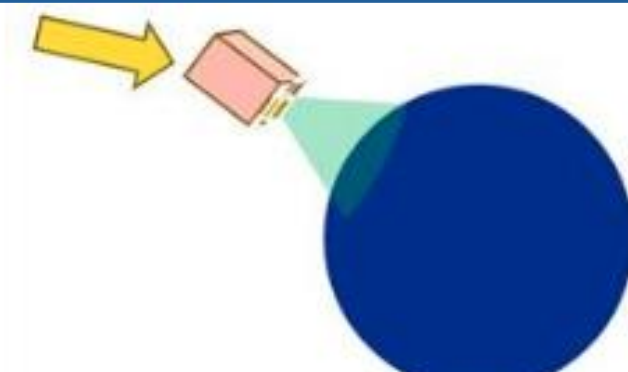
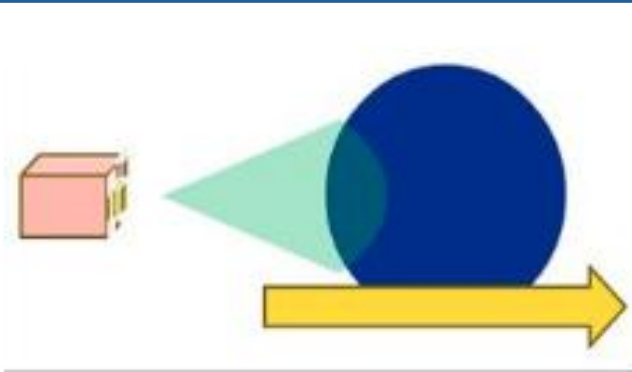
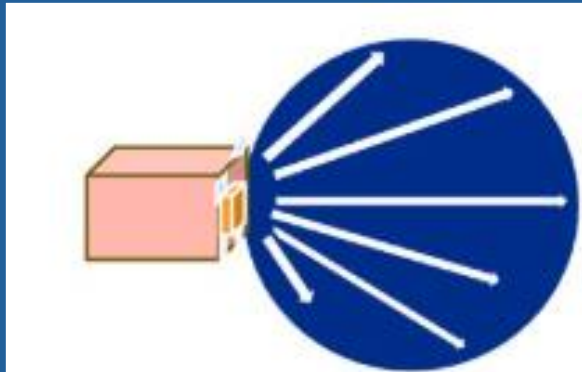
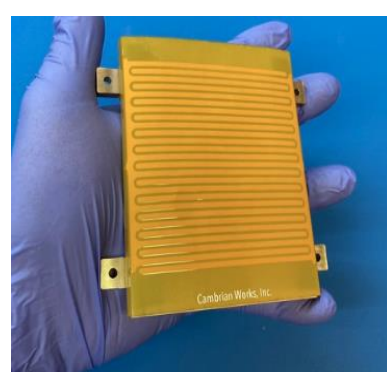
Mass: 20 kg (12U)

Carrying our own OBJECTIVEs (6)

6DOF Agile Maneuvering (Cold gas)

Stay within 10 meters

Payloads



eTAP

- This subsystem will provide the spacecraft with 1 eTAP pad.
- The eTAP Subsystem will be capable of capturing distance, force, and contact measurement data from on board sensors

Objective

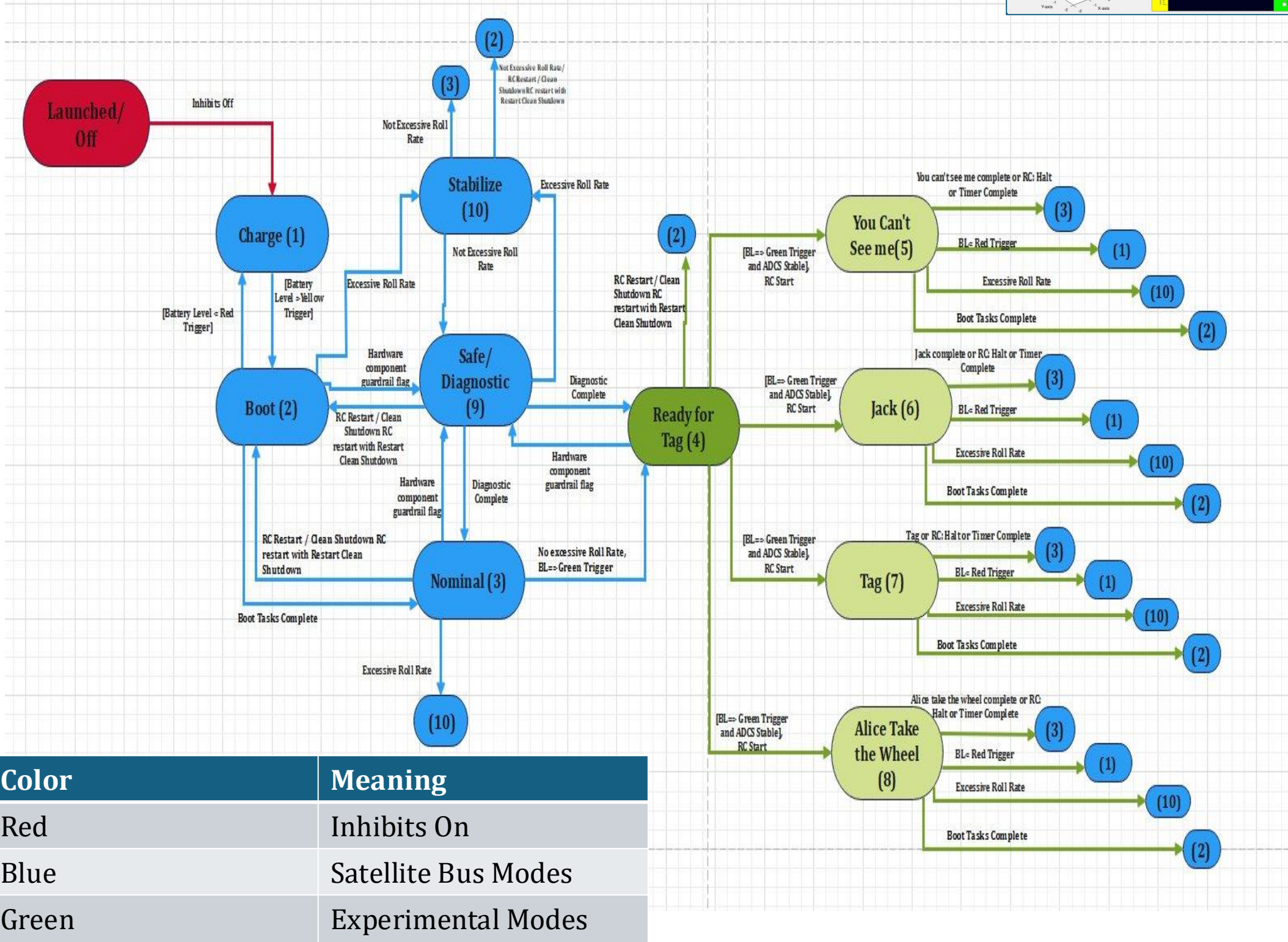
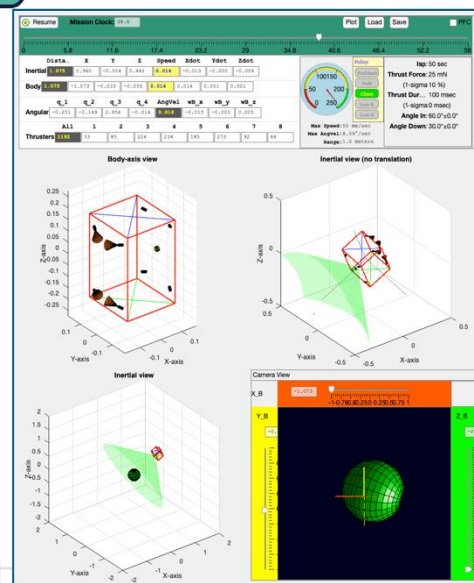
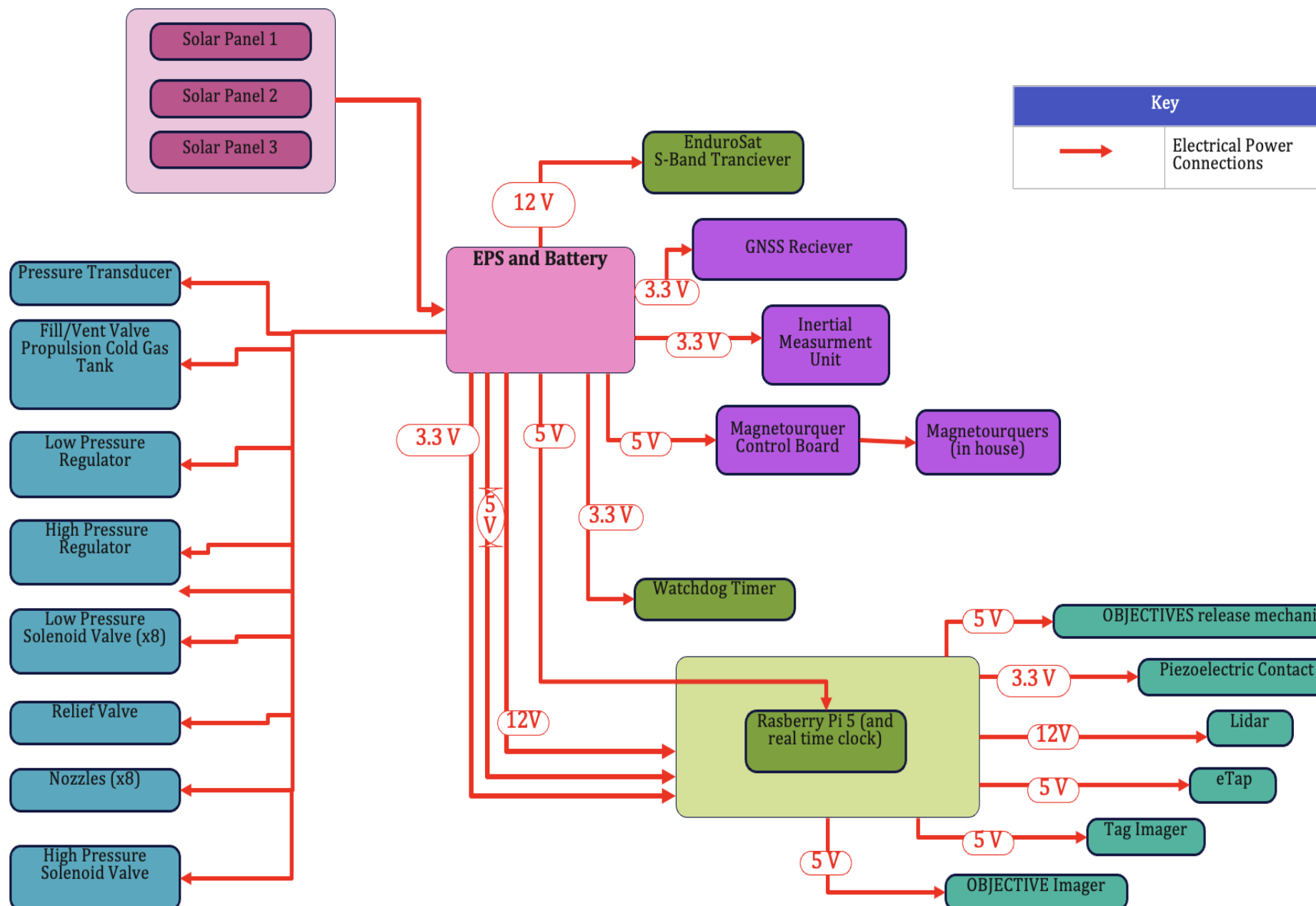
- The Objective Subsystem shall provide objectives of different surface materials
 - These will robust so that Sardonyx may hit them at a relative velocity without issue
 - These OBJECTIVEs will a surface measurable by the spacecraft's LiDAR
 - These OBJECTIVEs will be trackable by Space Force SSN radar
 - These OBJECTIVEs will deorbit in less than 5 years after release from the spacecraft

Maneuvering

- This subsystem shall have ability to know Sardonyx's orientation in space relative to an OBJECTIVE within 0.5(TBR) deg
- The Maneuvering Subsystem shall provide a propulsion system with cold gas thrusters and propellant
- The Maneuvering Subsystem shall provide thrusters that close and stay closed in the absence of power

Sensing

- The sensing subsystem will provide the spacecraft with a Lidar and other sensing devices
- The LiDAR shall have a field of view of at 90(TBR) degrees horizontally and 60 (TBR) degrees vertically
- The LiDAR shall be able to pick an Objective up to 10m away





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Space Systems Research Lab

Saint Louis University

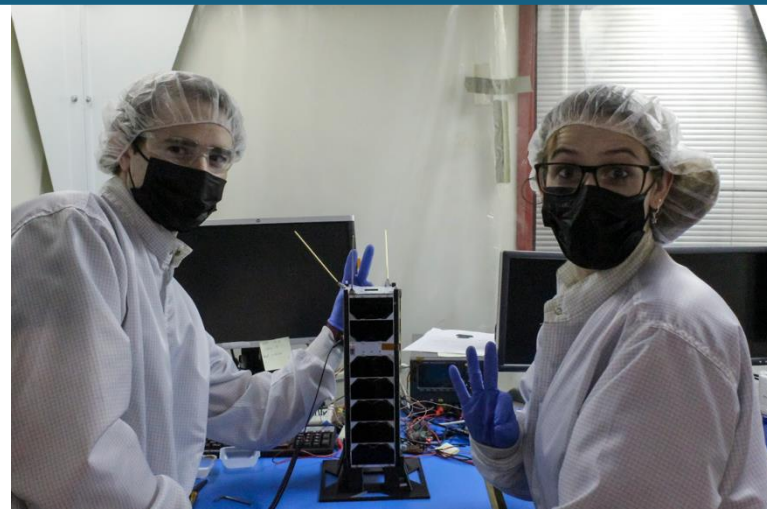
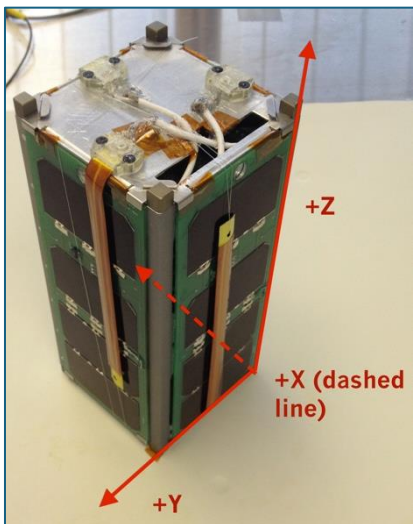
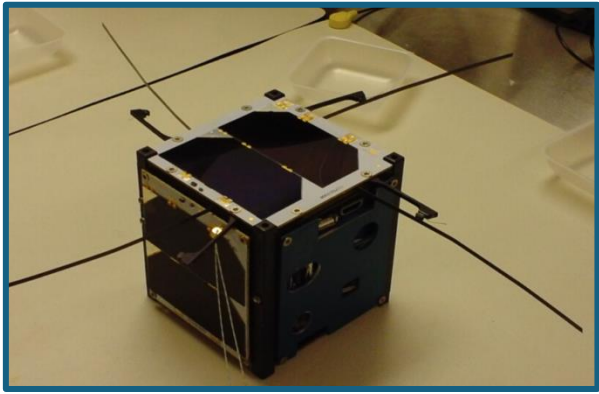
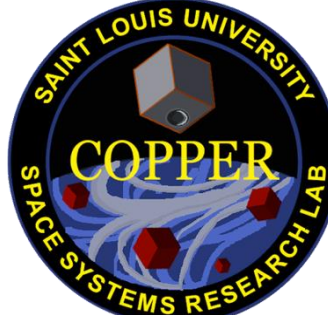
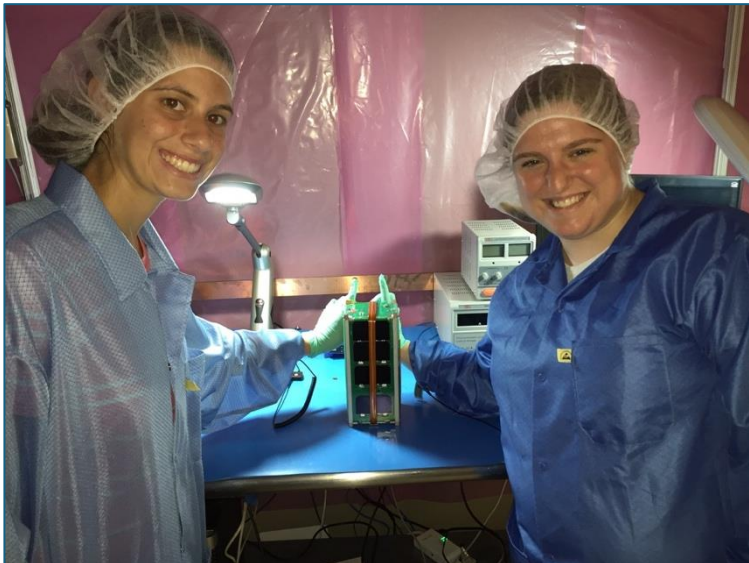
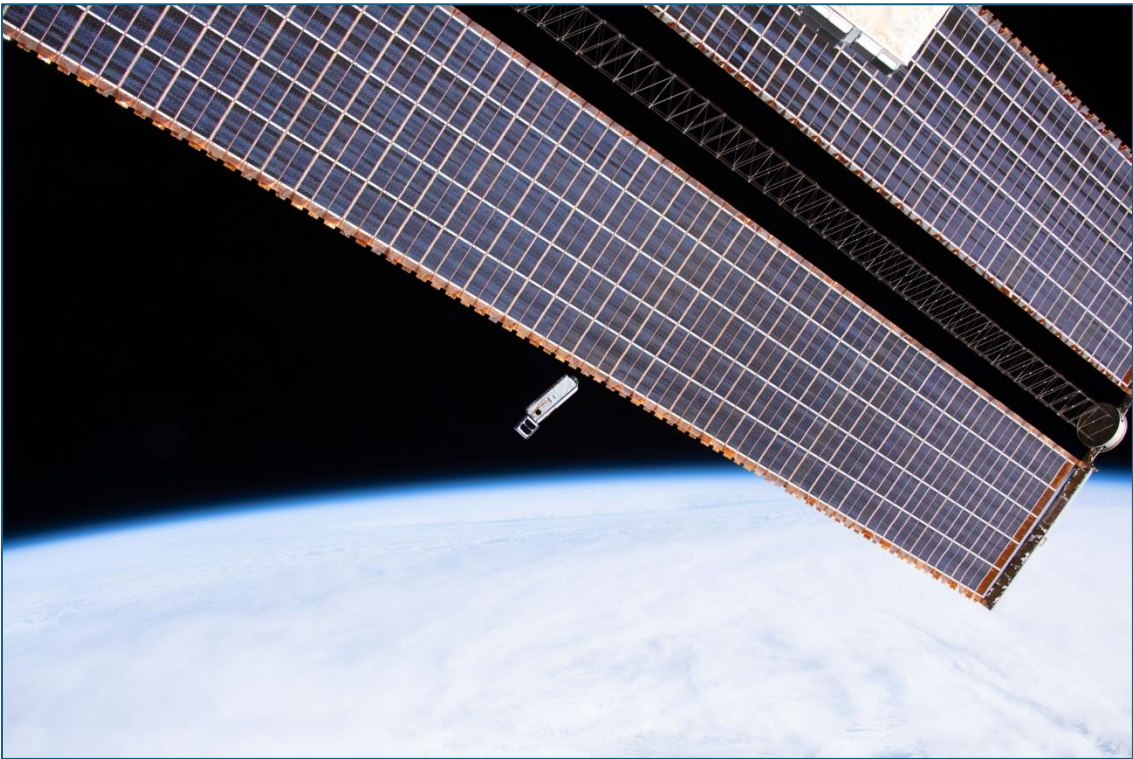
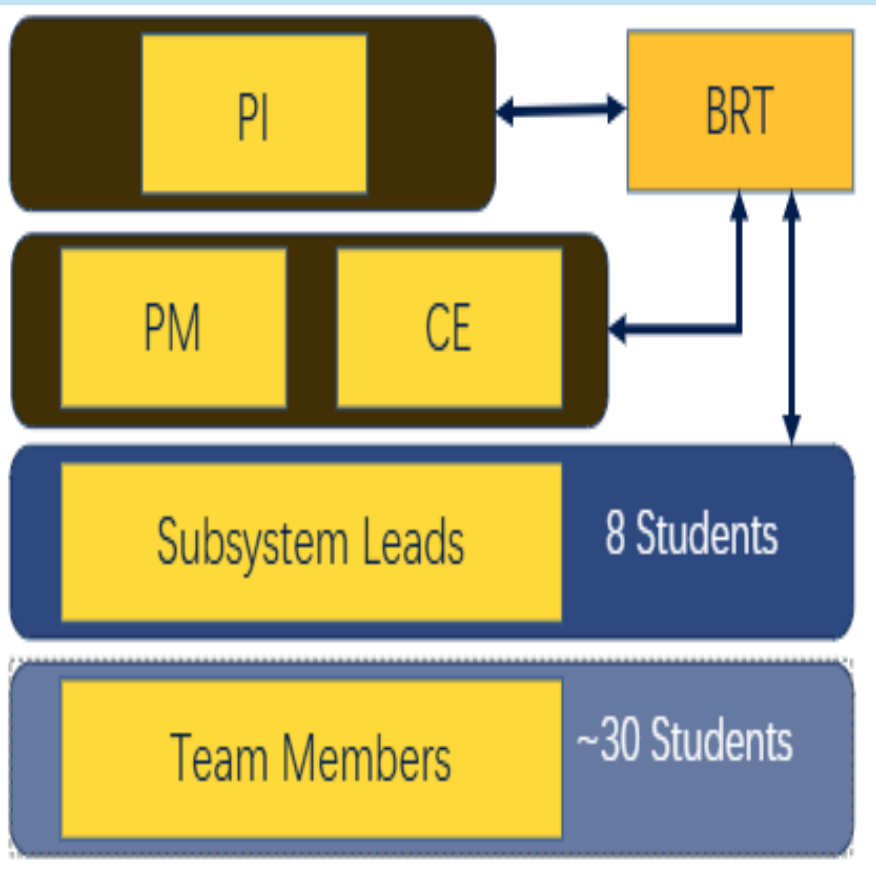


Space Systems Research Laboratory

- **Undergraduate focus:** training students in all aspects of space systems engineering by having students perform all aspects of space systems engineering
- **Graduate focus:** research in autonomy, mission assurance and proximity operations
- **Missions**
 - **COPPER** (SLU-01, NS-6, 2009-2011)
Launched: ORS-3/ELaNa-4 October 2013
 - **Argus** (SLU-02, NS-7, 2011-2013)
Launched: ORS-4/ELaNa-7 October 2015
 - **Argus-02** (SLU-03, 2018-2020)
Launched: ELaNa-21 October 2019
 - **DARLA** (NS-10, 2019-Present)
Launch Date: FR-1 October 2025
 - **DARLA-02** (NS-11, 2022-Present)
Launch Date: ???



Lab Organization



Flight Software (ARES)

Artificial Reasoning for Exploration and Space

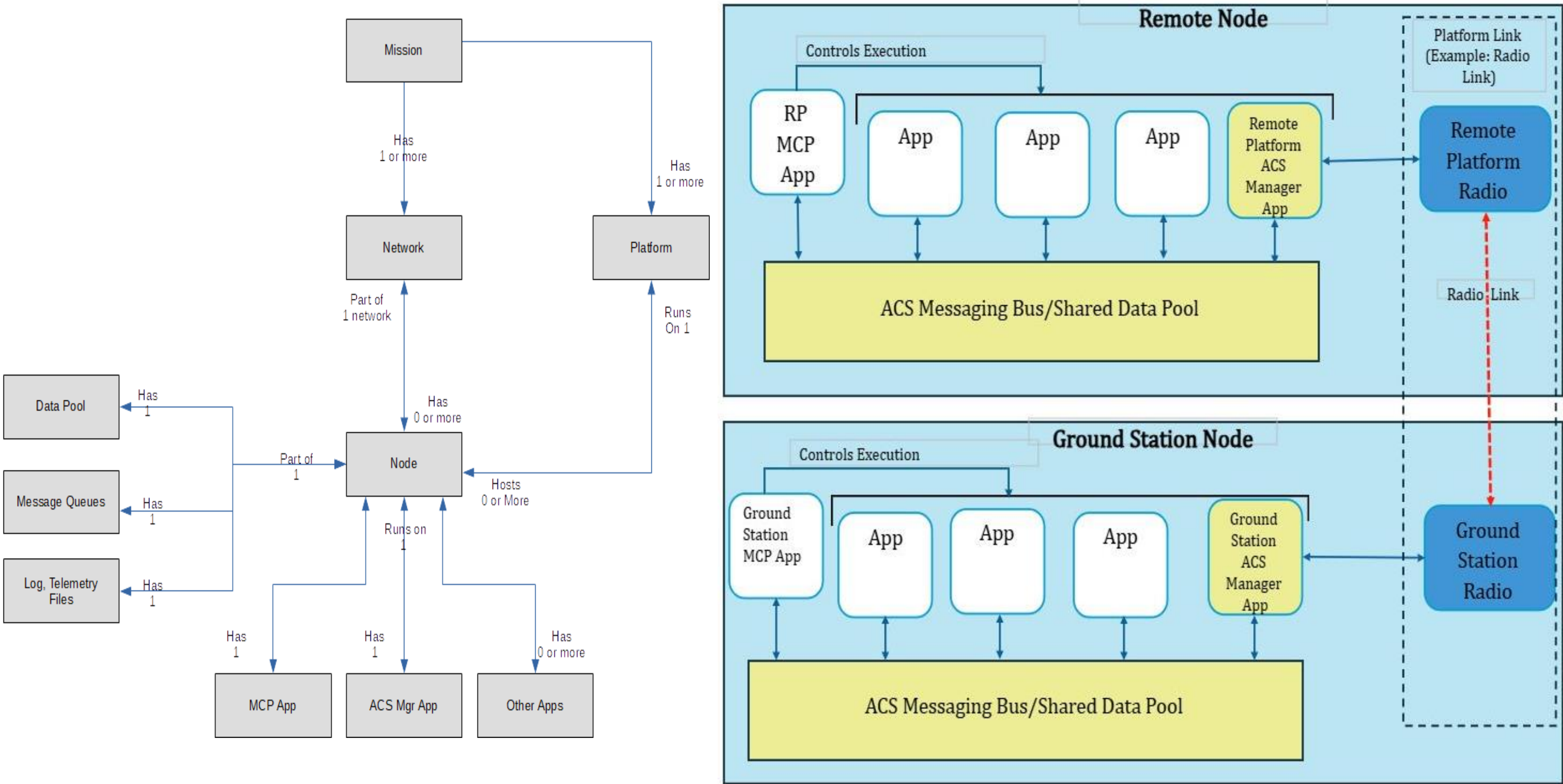
- ARES has been used and tested on many other missions prior to Sardonyx
 - DARLA-01 (integrated)
 - DARLA-02 (in progress)
 - ARGUS-02 (launched)
- ARES framework is comprehensive
 - Flight Software controls the execution of the Payload Software
 - The flight software also controls the execution of all spacecraft operations
- ALICE is an app within the ARES software that uses reinforced learning for Sardonyx's close proximity operations

ALICE APP

- Alice is a Sardonyx Specific App
 - Alice is a learning system to operate the Sardonyx thrusters to perform close approach and safe attachment to uncontrolled space objects
- The primary interest is to use flight data to calibrate the ground models and simulations, which would enable future, more complex flight operations
- It is believed that a well-trained Alice will outperform traditional relative-navigation methods when both are used to navigate around a tumbling shape that was not previously modeled

Long-term goal:

Learned Autonomy for spacecrafts, spacecraft constellations, and networks of spacecraft cooperating with ground resources



Baseline: Use SSRL’s DARLA-02 bus, with components replaced as needed

Sardonyx Bus

Structure

- 12U Tabbed form factor
- In-house design

Flight Processor

- Raspberry Pi 5
- In-house expansion board,
- ARES

Power

- Body-mounted cells (10 W average)
- COTS EPS / Battery

Attitude Determination & Control

- Magnetorquers for detumbling (not maneuvering)
- GNSS
- IMU, magnetometer

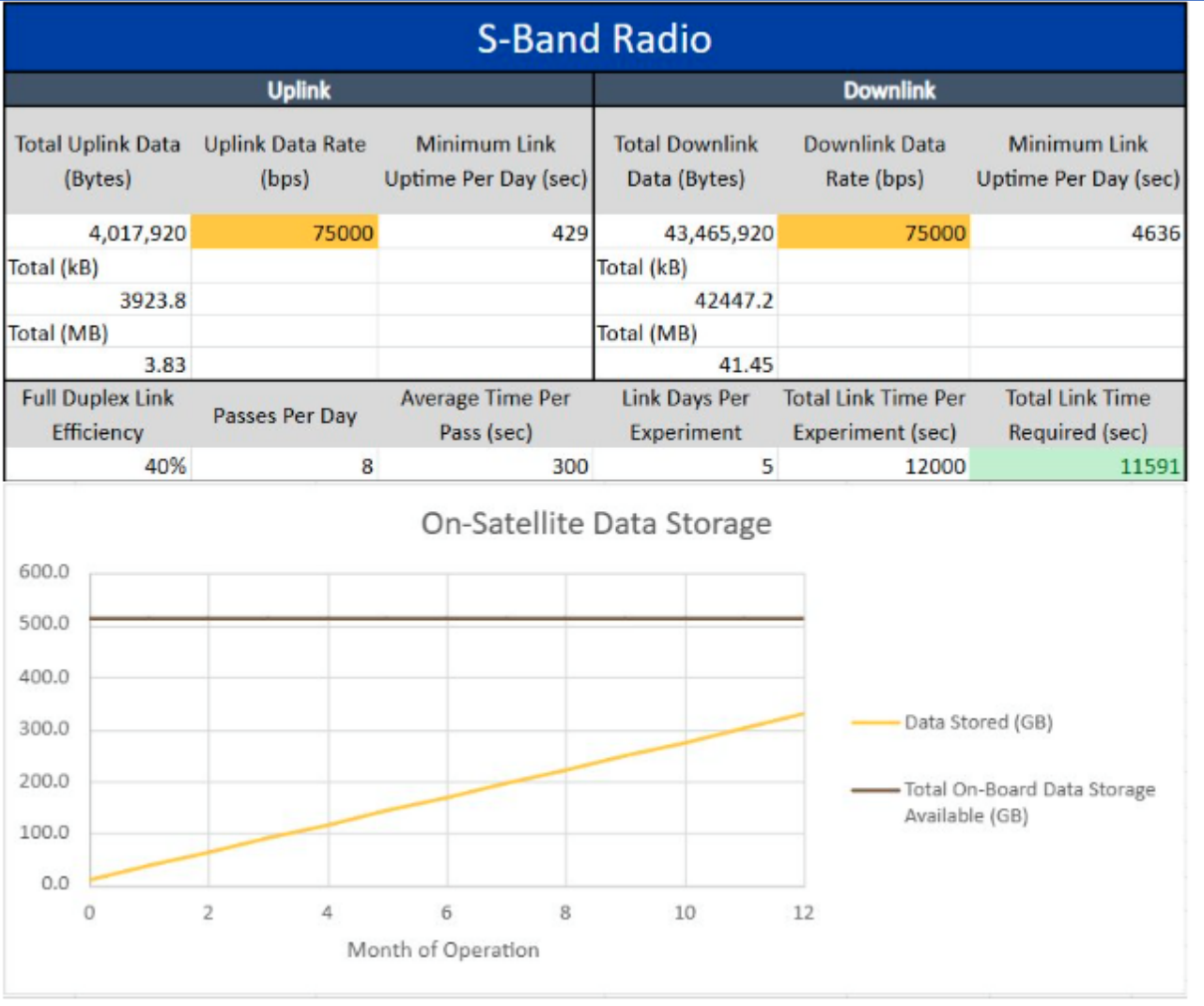
Communications

- S-Band Transceiver

Thermal

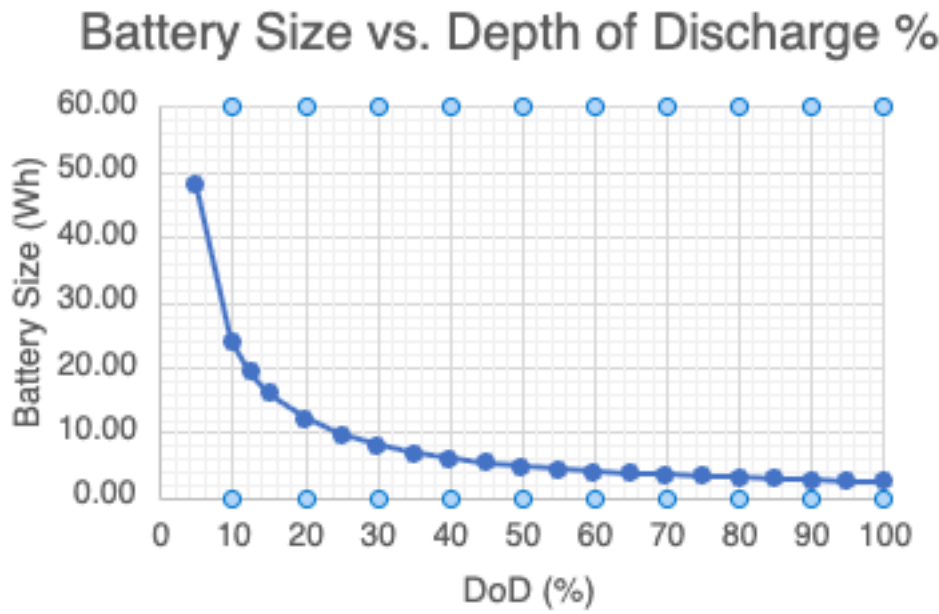
- Passive thermal control (TBR)

- 4 MB/week total uplink data
 - Ground commands
 - Alice updates
 - Uplink ack
 - Code uploads
- 45 MB/week downlink capacity (downlink over 5 days)
 - Lidar
 - Pi camera
 - Other range sensors
 - Alice
 - State of Health (all subsystems)
 - ARES overhead



Energy Generation	
Energy Generated over 24hr (Wh)	250.931
Average Power Generation (W)	10.455
Generated Power = 0.36 * Tag Power	
Generated Power = 1.17 * Nominal	

Energy Use By Subsystem		
	Tag Mode (W)	Nominal (W)
Flight Controllers	13.750	4.583
Sensing	7.859	0.000
Payload	0.586	0.130
Propulsion	3.171	0.211
ADC	0.680	0.818
Other	2.829	3.194
Total	28.874	8.937



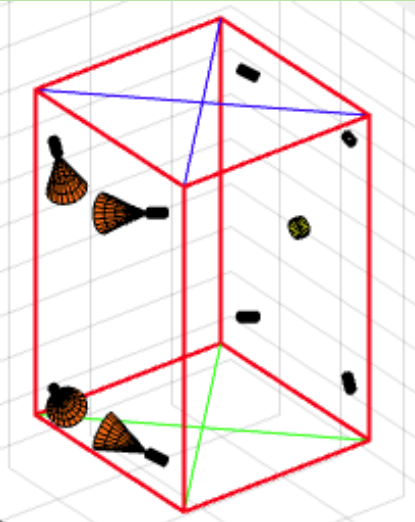
Subsystem	Total Subsystem Mass Including Contingency (g)	Mass Fraction (no Margin)	Mass Fraction (including Margin)
ADC	830	4%	3%
CDH	163	1%	1%
COM	533	3%	2%
MNV	4061	19%	17%
PLD	4038	19%	17%
PWR	4865	23%	20%
SNS	395	2%	2%
STR	6253	30%	26%
Baseline Total	21137	100%	88%
Maximum Allowable Mass	24000		
System Margin	2863	-	12%

All subsystem allocations include an internal 25% contingency

AMSAT/IARU Annotated Link Model System	Uplink	Downlink
Transmit Frequency (MHz)	2200	2100
Data Rate (bps)	100000	100000
Modulation	FSK	FSK
Transmit Power (W)	10	2
Transmit Power (dBm)	48.45	33.01
Transmit Gain (dBiC)	35	5
Path Loss (dB)	162.17	162.57
Receive Gain (dBiC)	0	18.9
Pointing, Polarization, Line Losses (dB)	14.1	3.41
Bit Error Rate	1.0E-5	1.0E-5
Eb/No	26.66	19.79
Eb/No Threshold	9.6	9.6
Margin	8.58	7.94

Sardonyx Hardware Mode Chart	Launched (0F)	Charging (1)	Boot (2)	Nominal (5)	Ready for Tag (4)	You can't see me (5)	Jack (6)	Tag (7)	Alice Take the Wheel (8)	Safe (Diagnostic) (9)	Stabilize (10)
EPS		x	x	x	x	x	x	x	x	x	x
Flight Computer			x	x	x	x	x	x	x	x	x
Transceiver				x	x		x	x		x	x
Magnetorquers				x	x					x	x
ADCS Sensors			x	x	x	x	x	x	x	x	x
eTap Pad (Tag)				(x)	x	x	x	x	x	(x)	
Objectives					x					(x)	
Objective Release System						x	x	x	x	(x)	
LIDAR					(x)	x	x	x	x	(x)	
Imagers				(x)	x	x	x	x	x	(x)	
Laser Rangefinders					x	x	x	x	x	(x)	
Contact Sensor					x	x	x	x	x	(x)	
Thruster Valves						x	x	x	x	(x)	
Notes:	x	(x)	The component is allowed to be active in this mode (and usually is)								
			The component is normally not active in this mode, but could be activated via ground command								

Maneuvering System



- 6DOF maneuvering via 8 thrusters
- Baseline Design: Parker Fluid Control Valves CO2 Propellant 250 grams = 5 m/s ΔV
- Robust potential-function-based controller

With great thanks to S. Pedrotty / NASA R5

Guardrails

Entering Safe Mode

- A valve is stuck open
- Components exceed temperature limits
- A component is malfunctioning
-
-
-
-
-
-
-
- NOT reasons to enter Safe Mode
- A clock reset

Abandoning a game of Tag

- More than 2500 (TBR) thruster firings
- OBJECTIVE is more than 10 meters (TBR) away
- Ground command can be sent to stop an experimental run
- The OBJECTIVE is lost for more than 300 seconds (TBR)
- The game exceeds 600 seconds (TBR)
- Entering Safe Mode (there is no automatic way back)
- Excessive roll rate
- Battery level has reached its red trigger
- NOT reasons to abandon a game of Tag
- A failed adhesion attempt